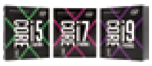
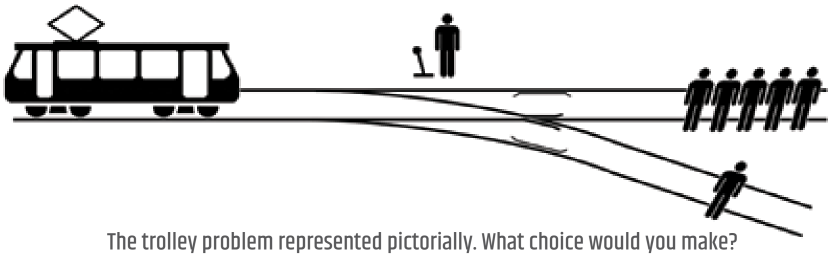


Intel's 18-core i9-7980XE
Intel's monstrous 18-core, 36-thread Core i9-7980XE CPU will launch on September 25 for a whopping $2000. **http://dgit.in/i97980XE**

DJI grounds Spark drones
DJI announces a new firmware fix For Spark and owners wont be flying their drones until they update it. **http://dgit.in/sparkDown**



The trolley problem represented pictorially. What choice would you make?

tracks, there are five people tied up and unable to move. The trolley is headed straight for them. You are standing some distance off in the train yard, next to a lever. If you pull this lever, the trolley will switch to a different set of tracks. However, you notice that there is one person on the side track. You have two options:

1. *Do nothing, and the trolley kills the five people on the main track.*
2. *Pull the lever, diverting the trolley onto the side track where it will kill one person. Which is the most ethical choice?*

This is where AVs cannot compete or surpass humans as of now. Human instinct would allow the human to factor in a number of things like the apparent age of people on either tracks, their social status etc - but most importantly, the value of life. If a vehicle, on a given day, is at all able to make the distinction between different people on the road, how can it be programmed to weigh in the value of life in a situation like this? The answer is not clear – not now, and doesn't seem likely to be in the near future either.

TO ROADKILL OR NOT TO ROADKILL?

One of the primary gaps in computer vision right now is the ability to understand what's inside an object. So, if a bag is suddenly thrown in front of a self-driving car, it will not possess the ability to understand if it's empty or is filled with bricks. The wrong last moment decision, in either case, could be fatal to the human passengers within.

While a human can clearly tell this is a stop sign, an autonomous car might fail

This brings us to one of the most simple, but problematic issues, with self-driving cars - sensor obstruction. Autonomous vehicles rely on their ability to see and make sense of things on the road to navigate - things like road signs. A recent study by the University of Washington shows that simple acts of vandalism on road signs, like adding a few dots and dashes to a Stop sign or making it read 'Love Stops Hate' was enough to confuse the onboard computer into assuming it as some other sign. In another case, a faded sign was not identified correctly. In either situation, it does not need to be explained how grim the outcome of either mistake could be in a live traffic situation.

THE WAY OUT?

To be honest, there are measures to deal with such situations. In fact, these kind of situations are the very reason that autonomous cars are being designed to accommodate inputs from multiple sources into their decision making process. For instance, hyper-accurate GPS mapping is used to correlate live imagery with map data on the roads. Endless hours of testing (which seems to be rare among many automobile companies that are into autonomous vehicles, according to numbers from California DMV) with humans involved can eventually make the on board computers on cars smart enough to judge human intuition. We could even go as far as assuming that with sufficient smart capabilities, non-autonomous vehicles will be able to communicate with autonomous vehicles on the road to understand their intentions and relay the information to their human drivers, hence avoiding any confusion brought by absence of eye-contact or body language. But that brings us to the last-straw - cybersecurity.

A VIRUS ON WHEELS

There is no evidence whatsoever right now to assure potential driverless-car owners that a computer onboard their vehicles will be any more secure than any other computer on a network. If you saw the Fate of the Furious earlier this year, you might have been sceptical about an entire city's cars going into auto-drive and wreaking havoc on the city. But then if you talked to a tech enthusiast or, even better, an automobile engineer, you would find out that it's not really an implausible situa-

That's not something you want to see on your dashboard

tion. While measures are being taken, every single one of us (and history) knows that there is more often a gap between regulation and action than not. Unfortunately, a miss here will not lead to just another batch of emails or accounts being leaked on the internet. It could lead to serious physical damage and even death.

WE NEED TO THINK

So before we jump onto the whole driverless bandwagon, automobile companies need to make sure that they are not rushing into something that they do not have the technology or ability to implement at its best possible state. Unless that's the case, we should all hold on to our horses and driverless cars should not be touted as the next promised revolution in transportation. And even if they are the next big thing, they are definitely way more than just a few years away. d